

From Ancient Symbols to Neural Networks: Charting the Evolution of Color and Frequency Mapping to Human Emotion

The mapping of sensory stimuli, particularly colors and sound frequencies, onto human emotional and cognitive states represents a long-standing inquiry at the intersection of psychology, neuroscience, art, and technology. This report provides a comprehensive exploration of the theoretical frameworks, models, and classification systems that define these relationships. It examines the foundational structures, from ancient philosophical traditions to modern computational models, and analyzes their psychological, technological, and symbolic implications. The investigation covers the mechanisms through which these mappings occur, the structured systems that enable empirical study, the real-world applications in fields like affective computing and therapy, and the emergent trends toward dynamic, personalized, and multi-modal systems. By synthesizing insights from diverse domains, this report aims to construct a holistic understanding of how sensory information is translated into subjective affective experience.

Foundational Frameworks: Philosophical, Cultural, and Cognitive Systems

The conceptualization of relationships between colors, frequencies, and emotions is not a recent development; it is deeply embedded in human history within symbolic, philosophical, and cultural frameworks. These systems provide a rich vocabulary and a structured worldview for understanding the affective power of sensory phenomena. They range from highly systematic, interconnected models like the Chinese Wu Xing theory to aesthetic classifications such as the Indian Rasa theory, and are further shaped by universal tendencies in human color categorization and linguistic relativity. These foundational systems form the bedrock upon which modern scientific inquiry builds,

providing both historical context and enduring principles that continue to influence contemporary thought and application.

One of the most intricate and holistic frameworks is the Chinese "Five Elements" or "Wu Xing" system [86 93](#). This ancient philosophy posits that all things in the universe are composed of five fundamental elements: Wood, Fire, Earth, Metal, and Water [168224](#). Each element is associated with a constellation of attributes, including a specific color, season, organ system, and emotion [226252](#). For instance, the element Fire corresponds to the color red, the season of summer, the heart and small intestine, and the emotion of joy [87 252](#). Conversely, the element Metal is linked to the color white, the season of autumn, the lungs and large intestine, and the emotion of grief [88 225](#). This system is not merely a static list of correspondences but a dynamic model of interaction governed by cycles of generation (nourishing) and control (constraining), aiming to achieve balance and harmony between humanity and nature [169](#). The philosophical underpinnings are rooted in Taoist doctrines, which are used to attract positive life energy in practices like feng shui [91](#). The symbolic values of these five colors were defined by Confucius during the pre-Qin period and related to virtues like benevolence [92](#). This integrated framework demonstrates a profound belief that environmental factors—such as color, sound, and season—are intrinsically linked to internal emotional and physiological states, offering a powerful model for understanding health and well-being through a lens of energetic correspondence.

Similarly, the Indian aesthetic tradition offers a sophisticated classification system known as Rasa Theory [294](#). Originating from the ancient treatise *Natyasastra* attributed to Bharata Muni, this theory provides a detailed framework for classifying aesthetic experiences [155157](#). Rasa, often translated as "flavor," "essence," or "sentiment," refers to the aesthetic emotion evoked in the audience by a performance [191192](#). The theory delineates nine primary Rasas, known as Navarasa, which represent core aesthetic feelings: love (*Sringara*), laughter (*Hasya*), sorrow (*Shoka*), anger (*Raudra*), heroism (*Vira*), fear (*Bhaya*), disgust (*Bibhatsa*), wonder (*Adbhuta*), and peace or tranquility (*Shanta*) [122270272](#). Each Rasa is not just an abstract emotion but is concretely symbolized by a specific deity, quality, and, crucially, a corresponding color [159160](#). For example, *Sringara*, the flavor of love and eroticism, is associated with the color light green, while *Raudra*, the sentiment of fury, is linked to the color red [123271](#). This system moves beyond simple emotion-color mapping to define a complex hierarchy of aesthetic feelings, deeply embedded in classical performing arts, visual arts, and literary criticism [294](#). The theory posits that the artist's skill lies in transforming transient, expressed emotions (*bhavas*) into these stable, aesthetic Rasas experienced by the spectator [191](#). The interplay of

dominant (*sthayibhava*), causative (*vibhava*), consequential (*anubhava*), and transitory (*vyabhicharibhava*) emotional states creates the rich tapestry of aesthetic experience that culminates in the emergence of Rasa ¹⁹³. This framework provides a culturally specific yet universally resonant model for understanding how art can systematically evoke and structure profound emotional journeys.

In contrast to these highly structured, non-arbitrary systems, Western frameworks have historically been more influenced by cultural conditioning and linguistic categorization. The work of Brent Berlin and Paul Kay on basic color terms provides a critical bridge between universal tendencies and cultural specificity ¹⁶⁶¹⁶⁷. Their research analyzed the color term systems of 98 languages and proposed a sequence of evolution for basic color categories, suggesting that certain color distinctions are more likely to be codified early in a language's development ¹²⁴¹⁸⁶. While they argued for commonalities across cultures that express universal features of human cognition, their findings also highlight the diversity of color lexicons ¹²⁷¹⁸⁵. This research supports the idea that while there may be a neurophysiological basis for perceiving certain color categories, the specific meanings and emotional connotations assigned to them are learned and culturally mediated ^{32 77}. This tension between potential biological universals and learned cultural interpretations is a central theme in the study of color symbolism ²⁵¹. The symbolic meanings of colors reveal wide-ranging connotations across different cultures; for example, red signifies luck and prosperity in some traditions, while white is associated with mourning in others ³⁰. This variability underscores the importance of context, culture, and individual demographics in shaping semantic associations ^{73 174}.

The commercial world has capitalized on these culturally derived associations through systems like the Pantone Matching System ¹⁸. The Pantone Color Institute functions as a hub for translating the "science and emotion of color" into actionable strategies for brands and designers ³⁰⁷³¹⁰. By uniting the technical precision of color matching with the psychological impact of color choice, Pantone provides services ranging from trend forecasting to brand identity development ³⁰⁸³¹². The selection of Pantone's "Color of the Year" is presented as a culmination of global trend analysis, capturing the collective mood and attitude of a moment in time ³¹³³¹⁵. This practice transforms a simple color into a cultural statement and a commercial directive, demonstrating the powerful role of color in shaping perceptions and promoting desired moods ³¹¹³¹⁶. Digital platforms like Pantone Connect allow designers to access production-ready colors directly within their design software, facilitating precise communication and reproduction of color choices intended to elicit specific emotional responses ¹⁴. This institutionalization of color

psychology reflects its widespread acceptance as a tool for influencing consumer behavior and brand perception.

Finally, cognitive models provide a more general, albeit less prescriptive, framework for understanding how humans process and categorize these relationships. One of the most influential models is the Arousal-Valence Model of Emotions [210](#). This model conceptualizes emotional experience along two primary dimensions: valence, which ranges from negative to positive, and arousal, which ranges from low to high [75](#) [263](#). Many studies investigating color-emotion associations plot responses within this two-dimensional space, allowing for a quantitative comparison of how different hues, saturations, and brightness levels affect emotional state [56](#) [232](#). For example, an achromatic scale from black to white can be mapped against these dimensions to show how changes in lightness correlate with feelings of relaxation or optimism [56](#). While this model provides a robust and widely accepted framework for describing emotional states, some researchers suggest it may not capture the full complexity of human affect, which might require additional dimensions to account for nuances like potency or dominance [217](#). Despite this, the Arousal-Valence model serves as a foundational coordinate system for empirical research, enabling the systematic mapping of how sensory inputs like color and sound translate into measurable emotional outputs. The existence of perceptual (dis)similarities between colors and emotions has been empirically demonstrated, indicating that these mappings are grounded in cognitive mechanisms that link the dimensional properties of sensory stimuli to the dimensional structure of feeling [55](#) [216](#).

Framework	Core Components	Associated Emotions/States	Key Principle
Wu Xing (Five Elements)	Wood, Fire, Earth, Metal, Water; Colors (Green, Red, Yellow, White, Black); Sounds; Seasons	Anger (Wood), Joy (Fire), Sympathy (Earth), Grief (Metal), Fear (Water) 225 252	Dynamic balance and cyclical interaction between elements.
Rasa Theory (Navarasa)	Nine aesthetic sentiments (Rasas); Deities; Qualities; Specific colors 159 160	Love (Light Green), Laughter (White), Sorrow (Grey), Anger (Red), Heroism (Light Orange), Fear, Disgust, Wonder, Peace 122 123	Transformation of transient emotions into stable, aesthetic experiences.
Berlin & Kay's Color Terms	Basic color categories (e.g., black, white, red, green, yellow) 126	Emotional associations are culturally learned and variable 174 250	Universal tendency for languages to evolve basic color terms in a predictable sequence 166 .
Arousal-Valence Model	Two orthogonal dimensions: Valence (positive-negative) and Arousal (calm-agitated) 75	Describes the position of any emotion within a 2D space (e.g., calm happiness, agitated anger) 232	Emotions can be quantitatively represented and compared based on these two core dimensions 263 .

These foundational frameworks, whether rooted in ancient philosophy, cultural practice, or modern psychology, collectively demonstrate a persistent human desire to find order and meaning in the relationship between the sensory world and the inner emotional life.

They provide the essential scaffolding for a deeper investigation into the mechanisms and applications of these mappings.

Neurophysiological Mechanisms Underpinning Color-Frequency-Emotion Mappings

While symbolic and cultural frameworks provide the "what" of color-emotion mappings, neurophysiological models attempt to explain the "how." These frameworks delve into the biological and cognitive processes of the brain and nervous system to elucidate the mechanisms that translate sensory input into affective and cognitive states. Central to this inquiry are theories of color vision, such as the trichromatic and opponent-process theories, which describe the initial stages of neural processing. Furthermore, the study of phenomena like synesthesia and cross-modal correspondence reveals how different sensory modalities interact and share neural substrates. Finally, the advent of brain-computer interfaces and frequency-based stimulation techniques like binaural beats allows for the direct modulation of brain activity, providing a tangible link between external frequencies and internal mental states.

The foundation of color perception itself is explained by two complementary theories: the Trichromatic Theory and the Opponent-Process Theory. The Trichromatic Theory, proposed by Thomas Young and later expanded by Hermann von Helmholtz, posits that human color vision is based on three types of cone photoreceptors in the retina, each sensitive to a different range of wavelengths corresponding roughly to short (S-cones, blue), medium (M-cones, green), and long (L-cones, red) wavelengths [206207](#). The combination of signals from these three cone types allows the brain to perceive the full spectrum of visible colors [208](#). This theory successfully explains phenomena like color mixing and the initial stage of photopic (daylight) vision. However, it fails to adequately explain certain perceptual phenomena, most notably color afterimages [207](#). When one stares at a colored image and then looks away, one perceives a complementary color (e.g., a green afterimage after staring at a red square). This suggests that color processing does not end at the cone level.

To address these shortcomings, Ewald Hering proposed the Opponent-Process Theory in the late 19th century [208209](#). This theory suggests that color information is processed in the retina and brain by neurons organized into three antagonistic systems: one for red vs. green, one for blue vs. yellow, and one for black vs. white (luminance) [206208](#). According

to this model, it is physiologically impossible to see a color as both reddish and greenish at the same time; the channels operate in opposition [209](#). The phenomenon of negative afterimages is elegantly explained by this theory: staring at a red stimulus fatigues the "red-on" cells, causing the unfatigued "green-on" cells to dominate when the gaze shifts to a neutral background, resulting in a perceived green afterimage [208](#). Modern neuroscience reconciles these two theories by showing that they operate at different stages of the visual pathway. The trichromatic signal from the cones is first processed by retinal ganglion cells that exhibit opponent responses, thus implementing Hering's mechanism at a neural level [207](#). While recent critiques question the strict validity of Hering's original model as a direct readout of hardwired neural mechanisms, the concept of opponency remains a cornerstone of visual neuroscience [205](#).

A profound parallel exists between the neurobiology of color processing and the neurobiology of emotion, stemming from the adaptation of Hering's opponent process theory by psychologist Richard Solomon in the 1970s [208](#). Solomon's theory of motivation and emotion posits that emotional states are also organized in opposing pairs (e.g., pleasure vs. pain, fear vs. relief) [208](#). He proposed that emotional reactions consist of an initial, intense positive or negative reaction followed by a secondary, longer-lasting reaction of the opposite valence. This model helps explain phenomena like addiction, where the initial pleasure from a drug decreases with repeated use, while the secondary withdrawal symptoms intensify, shifting the motivation from seeking pleasure to avoiding pain [208](#). This conceptual link is significant because it suggests a shared underlying principle of processing oppositional qualities, whether in the visual domain (red-green) or the emotional domain (pleasure-pain). The strong physiological basis for color opponency provides a plausible neurobiological substrate for why certain color combinations might evoke distinct emotional or cognitive states, even if the direct mapping is not explicit in Hering's original theory [206](#).

Beyond these foundational theories of vision, the study of synesthesia offers a unique window into the architecture of cross-modal perception. Synesthesia is a neurological condition where stimulation in one sensory or cognitive pathway leads to automatic, involuntary experiences in a second pathway [79](#) [99](#). Grapheme-color synesthesia, where letters or numbers involuntarily elicit the experience of specific colors, is one of the most studied forms [100](#). Research into its causes has led to two prominent neurocognitive models. The Cross-Activation Model, proposed by Ramachandran and Hubbard, suggests that synesthesia arises from atypical, increased neural connections (cross-wiring) between adjacent brain regions, such as the grapheme-recognition area and the color-processing area V4 [231256287](#). The Disinhibited Feedback Model proposes a different mechanism: in synesthetes, normal top-down inhibitory feedback from higher-level

cognitive areas to lower-level sensory areas is weakened, allowing signals to "leak" or spread uncontrollably between regions [227230](#). Evidence suggests that different types of synesthesia may involve different mechanisms, and some researchers propose a model that integrates both cross-activation and disinhibition, reconciling innate propensities with experiential influences [97 228](#). Synesthesia serves as a natural experiment, demonstrating that the boundaries between sensory modalities are not absolute and that the brain's wiring predisposes it to make connections across different domains of experience.

This phenomenon extends beyond the rare condition of synesthesia to a more common human tendency known as cross-modal correspondence. These are widespread, implicit associations that link features across different senses, such as associating high-pitched sounds with bright colors, high spatial positions, or small objects, and low-pitched sounds with dark colors, low positions, or large objects [82 83 84](#). These mappings are not arbitrary; they appear to be rooted in the shared neural substrates of different sensory modalities and are often mediated by emotion [40 108](#). Experimental evidence shows robust cross-modal matches between music and colors, and these matches are largely explained by the fact that both music and color can evoke similar emotional states [62 120](#). For instance, a happy, major-key melody is often matched with a bright, warm color, while a sad, minor-key melody is matched with a dark, cool color, because both the music and the color are conveying a similar sense of "happiness" or "sadness" [64](#). This emotional mediation provides a powerful psychological mechanism for how seemingly disparate sensory experiences become meaningfully connected in our perception. The existence of these correspondences has been demonstrated across various sensory pairings, including between pure vibrations and colors, linking vibrotactile stimuli to color and emotion [28 85](#).

Finally, technological interventions aim to leverage these neurophysiological principles to directly influence brain states through frequency-based stimulation. Brainwave Entrainment, most commonly practiced through Binaural Beats, is a technique where slightly different frequencies are presented to each ear [103](#). The brain perceives a third, phantom frequency equal to the difference between the two, and its electrical activity tends to synchronize with this beat frequency [107274](#). This is hypothesized to guide the brain into specific states associated with different frequency bands [44](#).

- **Delta (δ , <4 Hz):** Associated with deep, dreamless sleep and unconsciousness [106](#).
- **Theta (θ , 4-8 Hz):** Linked to drowsiness, meditation, creativity, and REM sleep [45 46](#).

- **Alpha (α , 8-12 Hz):** Characteristic of a relaxed, awake, and meditative state; often targeted for stress reduction [51](#) [267](#).
- **Beta (β , 13-30 Hz):** Corresponds to active, focused, and alert mental states; used to enhance concentration [267](#)[275](#).
- **Gamma (γ , >30 Hz):** Associated with higher-level cognitive functions like learning, memory, and information processing [49](#) [50](#).

Research on the efficacy of binaural beats presents mixed results. Some studies report positive effects on mood, anxiety, attention, and memory [50](#) [149](#). For example, gamma-frequency binaural beats have been shown to potentially enhance memory and attention [50](#), and alpha-band stimulation has been linked to reduced pain intensity [105](#). However, other rigorous reviews and experiments have found no significant changes in brain activity or emotional arousal [150](#)[262](#). The literature on working memory and gamma binaural beats, for instance, contains conflicting findings [260](#). This inconsistency highlights the need for more robust, methodologically sound research to establish clear causal links and practical applications. Nevertheless, the very existence of this technology represents a concerted effort to harness the brain's inherent oscillatory dynamics to modulate cognitive and affective states by manipulating external sensory input.

Structured Classification Systems for Empirical Research

The investigation of color-emotion and frequency-emotion mappings requires a high degree of precision and standardization to ensure that experimental conditions are controlled and results are replicable. This necessity has given rise to several structured classification systems designed to define, measure, and communicate color and light properties unambiguously. Among these, the Munsell Color Order System stands out as a particularly influential and scientifically rigorous framework, favored for its perceptual uniformity. Other systems, such as Pantone, NCS, and RGB, serve important roles in industry, digital media, and alternative perceptual models, providing a broader context for how color is categorized and utilized in research and application. The careful selection and application of these systems are paramount for isolating the specific effects of hue, value, and chroma on human emotional and cognitive responses.

Developed by American artist and educator Albert Henry Munsell in the early 20th century, the Munsell Color System is a three-dimensional model that organizes all visible

colors based on three perceptually uniform and independent attributes: Hue, Value, and Chroma [17](#) [20](#) . This was a significant departure from earlier color models, which often used regular geometric shapes like spheres or pyramids and were not based on human visual perception [20](#) . Munsell's key innovation was his focus on equal perceptual spacing, meaning the numerical steps he defined between colors corresponded to what the human eye perceives as an equal visual difference [17](#) . This scientific rigor makes the Munsell system exceptionally well-suited for controlled empirical research. The three dimensions are defined as follows:

- **Hue:** Refers to the dominant wavelength of a color, or what we typically think of as "color" (e.g., red, yellow, green) [13](#) [72](#) . Munsell established a hue circle divided into ten principal and intermediate hues: Red (R), Yellow-Red (YR), Yellow (Y), Green-Yellow (GY), Green (G), Blue-Green (PB), Blue (B), Purple-Blue (BP), Purple (P), and Red-Purple (RP) [21](#) .
- **Value:** Represents the lightness or darkness of a color, running along a vertical axis from black (Value 0) to white (Value 10) [20](#) [21](#) . The scale is divided into eleven equal intervals where the perceived difference between adjacent shades is constant [21](#) .
- **Chroma:** Denotes the purity or intensity of a color, measured radially from the center axis of neutral grays outward [20](#) [21](#) . A color with high chroma is vivid and saturated, while a color with low chroma is dull or desaturated. Munsell chose a step size of two units for chroma as it was found to be more easily perceptible than a single-unit step [21](#) .

Colors in the Munsell system are specified using a unique notation format (Hue/Value/Chroma), such as 5PB/4/8 for a moderately dark, bright purple-blue [21](#) . The Munsell Soil Color Charts, officially published in 1949, became a cornerstone of soil science, and the system is still used today in forensic pathology, dentistry, and other fields requiring precise color description [17](#) [20](#) . In the context of emotion research, numerous studies explicitly utilize calibrated Munsell hues to investigate the influence of specific colors on subjective and autonomic responses in immersive VR environments [2](#) [112136](#) . This allows researchers to isolate the effect of hue from other variables like saturation and brightness, providing a robust framework for testing hypotheses about color-emotion associations.

While the Munsell system is prized for its scientific basis, other classification systems are dominant in specific domains. The Pantone Matching System (PMS) is a proprietary color space system widely used in the graphic arts, printing, and manufacturing industries [18](#) [19](#) . Unlike Munsell, which is a perceptual model, Pantone is a commercial color reproduction system. Its purpose is to allow designers, printers, and manufacturers to

communicate and reproduce colors consistently and accurately across different materials and processes [18](#) . Colors in the Pantone system are created by mixing a set of 15 base pigments in varying amounts, ensuring a vast and precise palette that cannot always be replicated using standard CMYK (cyan, magenta, yellow, black) printing processes [19](#) . The Pantone Color Institute leverages this system to develop color trends and brand palettes, applying principles of color psychology to influence market perception and consumer behavior [307308](#) . Although primarily a tool for industrial and commercial communication, its widespread adoption means that many studies on color preference and emotional response use Pantone swatches to present color stimuli to participants [306](#) .

Another notable perceptual system is the Natural Color System (NCS), a proprietary model developed by the Scandinavian Color Institute [19 317](#) . The NCS is unique in that it is based on the color opponency hypothesis of color vision, first proposed by Ewald Hering [293](#) . It is considered a "perceptual" model because it describes color sensations as they are perceived in the upper brain, rather than at the receptor level like the RGB model [19](#) . The NCS is based on six elementary color percepts: white, black, red, yellow, green, and blue. All other colors are defined by specifying the amount of these six percepts present in a given sensation [19](#) . For example, a color is described by its degree of darkness, its chromaticness, and its percentage composition of two of four primary chromatic colors: red, yellow, green, and blue [306](#) . This approach aligns closely with Hering's opponent process theory and provides a different, though related, way of structuring the color space for research and design purposes [317](#) .

For digital displays, the RGB (Red, Green, Blue) additive color model is the standard. This model defines colors by combining different intensities of red, green, and blue light [19](#) . It is called "additive" because as more light is added, the color becomes lighter, eventually approaching white. The RGB model is the foundation for how colors are displayed on computer screens, televisions, and mobile devices. It is often used in conjunction with other models like HSV (Hue, Saturation, Value) or HSL (Hue, Saturation, Lightness), which are derived from the Munsell system's principles of separating hue from intensity and brightness [19](#) . In VR and AR research, where colors are generated digitally, the RGB model is the default system for specifying color stimuli [135](#) . However, because it is not perceptually uniform, researchers must be cautious when interpreting results based solely on RGB values, as equal numerical changes in RGB components do not always correspond to equal perceptual changes in color [17](#) .

The table below summarizes the key characteristics of these primary color classification systems:

System	Basis	Primary Dimensions	Key Feature	Common Application
Munsell	Perceptual Uniformity	Hue, Value (Lightness), Chroma (Intensity) 13	Based on experimental measurements of human visual response; equal steps are equally perceptible 17 20 .	Scientific research, soil science, forensics, academic settings 17 20 .
Pantone (PMS)	Commercial Matching	Pigment Mixtures	Proprietary system for consistent color reproduction across different materials and processes 18 19 .	Graphic design, printing, branding, manufacturing 18 308 .
NCS (Natural Color System)	Perceptual / Opponent Process	Darkness, Chromaticness, Percentage of 4 primaries (Red, Yellow, Green, Blue) 306	Based on Hering's opponent process theory; describes color as perceived in the brain 317 .	Design, architecture, industry in Scandinavia and Europe.
RGB (Additive)	Additive Light Mixing	Red, Green, Blue Intensity	Combines light to create colors; used for digital displays 19 .	Computer monitors, TVs, video games, VR/AR environments 135 .

The choice of a classification system is therefore not trivial; it dictates the nature of the research, the precision of the stimuli, and the interpretability of the results. The Munsell system's commitment to perceptual uniformity makes it a gold standard for controlled experiments seeking to understand the fundamental links between color properties and emotional responses. Meanwhile, systems like Pantone and RGB are indispensable for translating these findings into real-world applications in industry and digital media.

Technological Implementations in Affective Computing and Biofeedback

The theoretical frameworks and empirical findings regarding color and frequency mappings are increasingly being operationalized through advanced technologies, giving rise to the field of affective computing. This interdisciplinary domain focuses on creating systems that can recognize, interpret, and respond to human emotions [181292](#). These technologies are being deployed across a wide range of applications, from therapeutic interventions and educational tools to immersive entertainment and consumer wellness products. The implementation relies on a suite of sensing technologies to detect physiological correlates of emotion, followed by algorithms to classify these states, and finally, output mechanisms to deliver adaptive feedback. Key examples include electroencephalography (EEG) for decoding brain activity, remote photoplethysmography (rPPG) for unobtrusive vital sign monitoring, and consumer-facing biofeedback devices like the Reflect Orb.

Electroencephalography (EEG) is a prominent technology for emotion recognition due to its high temporal resolution and its ability to measure brain activity directly [7](#) . Different frequency bands within the EEG signal are associated with distinct cognitive and emotional states, providing a rich source of data for classification [106234](#). For instance, Alpha waves (8-13 Hz) are typically linked to a state of relaxed wakefulness, while Gamma waves (>30 Hz) are associated with higher-level cognitive functions like memory and information processing [45 49 267](#). Researchers leverage machine learning and deep learning techniques to decode emotional states from these signals. Studies have employed various classifiers, such as discriminative graph regularized extreme learning machines (GELM), to improve the accuracy of emotion recognition [9](#) . More advanced approaches involve complex neural network architectures like spatiotemporal graph transformers and multi-scale graph Mamba networks, which are designed to capture the intricate dynamics of EEG data across different subjects and sessions [116117](#). A significant challenge in this area is the high variability of EEG patterns between individuals, which affects the generalizability of models. To address this, researchers are developing domain adaptation techniques and attention mechanisms that can automatically learn the most relevant EEG channels and frequency bands for a given person [12](#) . The ultimate goal is to build robust, subject-independent emotion classification systems that can be used in real-world scenarios [117](#).

Remote Photoplethysmography (rPPG) represents a paradigm shift towards unobtrusive emotion sensing. This optical technique uses a standard camera to measure minute variations in skin color caused by changes in blood volume beneath the skin's surface [96 202](#). As the heart pumps blood, the slight increase in dermal perfusion alters the skin's reflectance of light, which can be captured by an RGB camera and analyzed to extract vital signs like heart rate and heart rate variability (HRV) [279291](#). HRV, in particular, is a valuable indicator of autonomic nervous system balance and is closely linked to emotional states like stress and relaxation [246](#). The primary advantage of rPPG is its contact-free nature, making it ideal for continuous monitoring in naturalistic settings. However, the technology faces significant challenges. Its accuracy is highly sensitive to ambient lighting conditions, motion artifacts from the user, and critically, the pigmentation of the user's skin [94 95](#) . Research has shown that melanin in darker skin tones absorbs more light, leading to a weaker signal and potential measurement bias [239 240](#). This has resulted in documented racial biases in pulse oximetry and rPPG devices, raising serious ethical concerns about equitable healthcare technology [236237](#). To mitigate these issues, researchers are exploring multi-spectral approaches that incorporate near-infrared (NIR) light, which is less affected by melanin, to improve the reliability of rPPG across diverse populations [201](#).

A prime example of a consumer-facing technology that applies these principles is the Reflect Orb, a biofeedback device designed to help users manage stress and cultivate calm [197200](#). Developed by the Israeli startup Reflect Innovation, the device consists of a handheld orb that the user holds, making contact with sensors on its surface [199](#). These sensors measure lab-grade physiological signals from the fingertips, primarily Electrodermal Activity (EDA), which detects stress through sweat gland activity, and Electrocardiogram (ECG), which measures heart rate [195245](#). The device translates this raw physiological data into a simple, intuitive visual cue: a ring of light at the top of the orb that shifts color in real-time [197](#). The color progression typically moves from purple or red to signify high stress, through white or yellow for neutral, to calming blue or white for a relaxed state [5](#) [194](#). This immediate visual feedback serves as a "facilitator of learning," allowing users to see the direct impact of their own actions, such as slow breathing, on their nervous system's state [199243](#). The device is paired with a mobile app that provides guided practices and tracks progress over time [195](#). Scientific validation for the device includes randomized controlled trials that showed significant improvements in stress, anxiety, depression, and sleep quality among users after a four-week period [196198](#). Following the October 7 Hamas massacre in Israel, the company pivoted its focus towards trauma therapy, collaborating with medical organizations to use the device for treating survivors, highlighting its transition from a wellness product to a clinical therapeutic tool [199244](#).

These sensing and interpretation technologies form the basis of adaptive and responsive environments, particularly in the realm of Virtual Reality (VR). VR offers a controlled yet immersive platform for eliciting and regulating emotions [135](#). Studies have demonstrated that simply changing the color of virtual room walls can significantly alter a user's emotional state and physiological arousal [247301](#). Building on this, more advanced systems are being developed to create closed-loop, real-time adaptive environments. The "VEE-loop" (Virtual-Reality-Based Emotion-Elicitation-and-Recognition loop) is a proposed framework for integrating emotion elicitation, recognition, and environmental adaptation within immersive VR [1](#) [277](#). In such systems, a user's emotional state is continuously monitored using biosensors like EEG or respiratory trackers. If the system detects signs of heightened anxiety, for instance, it can dynamically adjust the virtual environment—for example, by softening the lighting, playing calming music, or changing the scenery—to help regulate the user's emotional response [68](#) [249](#). One study examined a real-time adaptive system using breathing rate as a biofeedback visual cue in an immersive forest environment to support emotional regulation [68](#). Another project, NEUROWOODS, investigated a virtual forest environment driven by real-time EEG feedback to enhance emotional regulation and cognitive benefits [249](#). These systems move beyond passive

exposure to active, personalized intervention, leveraging the power of immersion to create truly affective computing experiences.

Technology	Principle of Operation	Target Signals	Applications	Challenges
EEG	Measures electrical activity of the brain via scalp electrodes.	Brainwave frequency bands (Delta, Theta, Alpha, Beta, Gamma) 106	Emotion recognition, cognitive state monitoring, brain-computer interfaces 6 7	High inter-subject variability, sensitivity to noise, requires expert setup.
rPPG	Uses a camera to analyze subtle changes in skin color from reflected light 202 .	Heart Rate, Heart Rate Variability (HRV) 96	Unobtrusive vital sign monitoring, contactless emotion sensing 94	Sensitive to lighting, motion, and skin pigmentation bias 237 .
Biofeedback Devices (e.g., Reflect Orb)	Measures physiological signals (EDA, ECG) and provides real-time visual feedback 195 197 .	Skin Conductance (EDA/GSR), Heart Rate (ECG) 245	Stress management, anxiety reduction, self-regulation training, trauma therapy 199	Requires user engagement and practice; effectiveness varies by individual.
Adaptive VR	Integrates real-time physiological feedback to dynamically alter the virtual environment 1 .	Multiple (Heart Rate, EEG, Breathing, Eye-tracking) 3 70	Exposure therapy, emotional regulation, pain distraction, mental wellness 132 249	High development cost, potential for cybersickness, requires accurate sensing.

Collectively, these technologies illustrate a clear trajectory from passive observation to active intervention. By embedding sensing and feedback mechanisms into everyday objects and immersive environments, they enable a new form of human-computer interaction where technology is not just a tool to be used, but a partner in managing our own emotional well-being.

Emerging Paradigms: From Static Associations to Dynamic, Multi-Modal Systems

The field of color and frequency mapping is undergoing a significant evolution, shifting from simplistic, static, one-to-one associations toward complex, dynamic, and multi-modal systems. This paradigm shift reflects a deeper understanding of the multifaceted nature of human emotion and cognition. Early research often sought to define universal mappings, such as "red equals anger" or "high pitch equals bright color." While these foundational observations remain valid in certain contexts, contemporary frameworks recognize that emotional responses are highly contextual, individualized, and emerge from the integration of multiple sensory streams. The emerging trends point toward systems that are adaptive, personalized, and capable of processing and generating affective experiences across auditory, visual, and even haptic modalities.

A key trend is the move from static associations to dynamic adaptation. The concept of "adaptive color" exemplifies this shift, where an interface or environment adjusts its color palette in real-time based on a user's inferred emotional state [212](#). This goes beyond the traditional use of color to evoke a specific feeling and instead creates a responsive dialogue between the user and the system. For example, an adaptive learning environment could become more vibrant and stimulating if it detects boredom, or it could shift to cooler, calmer colors if it senses frustration [181](#). This dynamic adjustment is made possible by the integration of emotion recognition technologies. The Music2Palette framework is a compelling example of generative adaptive systems; it uses cross-modal representation learning to analyze the emotional content of a piece of music and then generates a corresponding color palette that aligns with that emotion [39](#). Similarly, AI-driven systems like EmoPalette can recognize emotions from facial expressions and generate an adaptive color palette in real-time [290](#). These systems represent a move from prescribing a fixed emotional response to co-creating an evolving affective experience with the user.

This dynamism is further enhanced by the development of closed-loop biofeedback systems. These systems form a complete circuit: they sense a physiological state, interpret it as an emotional state, and then deliver a tailored sensory feedback signal to guide the user toward a desired state. The Reflect Orb is a consumable embodiment of this principle, translating EDA and ECG signals into a changing color ring to facilitate self-regulation [194197](#). On a more technologically advanced front, wearable systems like Emotion-to-Motion translate physiological signals from EEG and GSR into real-time visual and kinetic feedback, providing another channel for users to engage with their internal states [138](#). In VR, closed-loop systems integrate real-time feedback loops that respond dynamically to users' emotional and physiological cues, such as heart rate and skin conductance, to create personalized therapeutic experiences [211](#). For instance, a system might adjust the pace of a VR narrative or the intensity of a virtual exposure scenario based on the user's galvanic skin response, effectively tailoring the intervention to their unique emotional threshold [248](#). This closed-loop approach marks a significant advancement from open-loop systems, which simply present a stimulus without adapting to the user's response.

Another critical direction is the integration of multiple sensory modalities. Human emotion is rarely experienced through a single sense in isolation; it is a multisensory phenomenon. Therefore, the most effective affective systems are beginning to move beyond purely visual or auditory channels to embrace cross-modal correspondences. Research has shown that audiovisual emotional congruency enhances affective experiences, and that cross-modal interactions can shape how emotions are perceived [63](#)

254. The DHGTN (Dynamic Heterogeneous Graph Temporal Network) is an end-to-end framework designed specifically to model these dynamic cross-modal interactions, effectively learning how information from different sensory inputs (like audio, video, and text) combines to produce an emotional output 41. Other models, such as the Graph-Attention Framework for Multimodal Emotion Recognition, use heterogeneous graphs to capture the semantic relations between different modalities, allowing the system to weigh the importance of each input dynamically 130. Wearable prototypes have been developed that integrate visual and auditory feedback with real-time respiratory biofeedback to support rapid emotional regulation 69. This multi-modal approach acknowledges that the brain processes affective information holistically, and that leveraging correspondences between, for instance, music and color, can create a more powerful and coherent emotional impact than either modality alone 40 108.

Furthermore, there is a growing emphasis on personalization and overcoming the limitations of one-size-fits-all models. Individual differences in emotional perception are substantial and are influenced by factors like artistic training, gender, age, and cultural background 16 110174. A study comparing art students to non-art students found that while both groups responded to color brightness and saturation, art students were more sensitive to hue, whereas non-art students were more strongly affected by saturation 16. This highlights the need for systems that can adapt to an individual's specific sensitivities. Personalized binaural beat protocols, for example, are being explored, with research suggesting that the clinical efficacy of gamma entrainment may depend on an individual's unique neurophysiological receptivity 215. The future of affective computing likely lies in building hierarchical frameworks that can estimate high-level emotional judgments while locally integrating this information to adapt to a user's unique profile 42. This involves moving from population-level averages to individual-specific models, potentially using techniques like transfer learning and personalized machine learning to tailor interventions for maximum effectiveness 238.

Finally, the hierarchy of these systems appears to be evolving in layers. At the most basic level, there are automatic, potentially universal, responses to fundamental color properties. For example, research consistently shows that higher brightness and saturation are generally associated with higher arousal, regardless of hue 153300. This is supported by findings that the most critical factors in the affective meaning of color are brightness and saturation, with hue having a more secondary role 302. At a higher cognitive layer, these basic responses are filtered and modulated by learned associations, cultural context, and individual experience 73 306. A color that is arousing in one culture may be neutral or even aversive in another. At the highest level, these modulated

responses are integrated with other sensory inputs within a complex, dynamic system. The Visualized Emotion Ontology (VEO), for example, attempts to model emotions using a combination of shapes, lines, and colors, representing a move towards a more holistic, multi-parameter representation of affect [114](#). This layered processing model—from automatic physiological responses to cognitively-mediated interpretations to integrated multi-modal experiences—provides a more nuanced and realistic framework for understanding and designing affective systems. It acknowledges that emotion is not a simple input-output relationship but a complex, ongoing process of interaction between the body, the brain, and the environment.

Synthesis of Implications and Future Directions

The extensive body of research on frameworks that map colors and frequencies to emotional and cognitive states yields profound implications across psychological, technological, and symbolic domains. The synthesis of these findings reveals a field characterized by a convergence of ancient wisdom and cutting-edge technology, unified by a fundamental quest to understand and harness the deep-seated connections between our senses and our inner lives. The journey from Munsell's perceptually uniform color charts to AI-driven, closed-loop biofeedback systems illustrates a powerful evolution in methodology and ambition. This final analysis distills the key implications of these mappings and outlines the critical questions and directions that will shape the future of this interdisciplinary endeavor.

From a psychological perspective, the research confirms a robust, though nuanced, link between sensory stimuli and affective states. There appears to be a universal foundation for certain color-emotion associations, such as the tendency for warm colors to be linked with higher arousal and for saturated colors to evoke stronger emotional responses [153250300](#). The Arousal-Valence model provides a useful universal metric for quantifying these responses, suggesting a common dimensional structure to emotional experience across individuals [75 184](#). However, this universal foundation is significantly modulated and complicated by a host of cultural, linguistic, and individual factors [78 174](#). The work of Berlin and Kay on color term universality versus cultural relativism provides a powerful analogy for this dynamic; while the brain may be predisposed to certain perceptual groupings, the specific emotional labels and meanings attached to them are learned and culturally contingent [32 127](#). Furthermore, individual differences are substantial, with factors like artistic training, gender, and age demonstrably influencing emotional

perception [16 110](#). This necessitates a move away from monolithic, one-size-fits-all models toward personalized and context-aware frameworks. The parallel between Hering's opponent process theory of color vision and Solomon's theory of emotion is a particularly insightful finding, suggesting a shared neurobiological substrate for processing oppositional qualities, whether in the visual spectrum or the emotional landscape [208](#).

Technologically, the implications are far-reaching, promising a future where our environments and devices are acutely aware of and responsive to our emotional needs. The enabling technologies—advances in EEG, rPPG, AI, and VR—are creating unprecedented opportunities for affective computing [7 96 181](#). Applications are already emerging in mental health, where adaptive VR systems are being used for anxiety regulation and exposure therapy [132249](#), and in consumer wellness, exemplified by biofeedback devices like the Reflect Orb that empower users to manage their stress response [195197](#). The potential for these technologies in education, marketing, and entertainment is immense, offering the ability to create deeply engaging and emotionally resonant experiences [181315](#). However, this technological promise is accompanied by significant challenges and ethical considerations. The documented bias of rPPG technology against individuals with darker skin tones is a critical issue of equity and social justice that must be addressed to prevent the reinforcement of existing health disparities [237239](#). The inconsistent scientific evidence for the efficacy of technologies like binaural beats serves as a reminder of the need for rigorous validation before widespread deployment [150260](#). Moreover, as these systems become more adept at inferring and influencing our emotional states, profound questions about privacy, manipulation, and autonomy arise. Who controls the "emotional state" of a user in an adaptive environment? How is sensitive physiological data collected, stored, and used? These are pressing ethical dilemmas that demand thoughtful consideration and regulation.

Symbolically and for design, the traditional frameworks offer a timeless vocabulary and conceptual toolkit. Understanding that hue, saturation, and lightness have differential impacts on emotion allows for more intentional and effective design in fields ranging from interior design and fashion to web design and branding [16 173](#). Warm, saturated, and bright colors are understood to be energizing, while cool, desaturated colors are seen as calming [153](#). Artistic frameworks like Cugelman's Emotion Map or the Navarasa theory provide artists and designers with structured ways to navigate and depict the complex emotional journey of a narrative or experience [59 159](#). Color functions as a potent symbolic signifier, capable of communicating messages and emotions effectively without words [113115](#). This semiotic function is leveraged commercially by institutions like the

Pantone Color Institute, which translates cultural zeitgeists into strategic color palettes that brands can use to shape perception and drive consumer behavior [307312](#).

Looking forward, several key gaps and future directions emerge from this analysis. First, while correlations are well-established, proving direct causality between a specific sensory stimulus and a specific emotional outcome remains a formidable challenge due to the myriad of confounding variables inherent in human experience. Second, the majority of studies focus on acute, short-term emotional responses. The long-term psychological and behavioral effects of living in or regularly interacting with these adaptive, affective systems are largely unknown and warrant long-term longitudinal research. Third, the integration of ancient, holistic systems like Wu Xing and Rasa theory with modern, data-driven computational models represents a significant opportunity. Could a hybrid system, informed by both centuries of cultural wisdom and granular neuroscientific data, lead to more effective, ethically grounded, and holistic applications in health and well-being? Finally, as these technologies mature, fostering public literacy and participation in their development will be crucial to ensure they serve the public good and align with societal values. The future of this field lies not just in building more sophisticated algorithms, but in cultivating a deeper, more responsible understanding of the profound connection between our senses, our brains, and our humanity.

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